Platform Domains

The domains outlined below are the result of integrating the initial design with Domains_Combined_v3 and new requirements. This incorporates relevant patterns and structures to strengthen the overall model.

Overview

1 · Authoritative — hold legal, financial & enforcement authority

Claims & Reimbursement · Accounting & Disbursement · Compliance & Review · Eligibility & Direct Certification

AUTHORITY BOUNDARY — nothing below decides money, eligibility, or enforcement

2 · Foundational — constrain inputs; never decide outcomes

Identity & Access · Configuration & Tenant · Sponsor & Site Registry · Applications & Agreements · Program Registry (DATA SPINE)

SPECIALIST BOUNDARY — business-critical, not authoritative

3 · Operational — own program rules & integration surfaces

Programs: SNP · CACFP · DCH · SFSP · FFVP · SSFO · CEP Platform services: Forms Engine · Workflow & Task Mgmt · Integration & Data Exchange · Document Library

OBSERVATIONAL BOUNDARY — downstream consumes facts; never influences decisions above

4 · Observational — read models & outbound services only

Reporting & Analytics (→ Snowflake) · Audit & Observability · Notifications · Communications · Reference Data & Transparency · LaunchDarkly

cnp_global (1 shared Aurora instance — 387 tables: rates, types, codes, templates) + cnp_ops (1 Aurora per state — 1,015 operational tables). Database IS the tenant boundary. Annual rates = INSERT with EffectiveDate. Zero SP changes.

Tier 1 · First-Class Authoritative Domains: Legal, financial & enforcement authority

Claims & Reimbursement

Facts of service. Federal math. Never money.

Mal counts. Claim validation (federal SPs → C# with characterization tests). Effective-dated ClaimLaws. Advance requests. Claim locking. Non-congregate 3-mode validation. Section 4/11 split for FNS-10. Auto-Decline Claims (ME pattern — configurable per tenant via feature flag).

Accounting & Disbursement

Federal grants. State money flowing down.

Grant tracking. DOF calculation. FNS reports (10/44/418/640/905). Repayment schedules. IAccounting Exporter per state. CN Financials. Earnings/269. AFR Sponsor Reporting (Annual Financial Report — IN center/DCH variants; ME NSLP variant — both are reference implementations).

Compliance & Administrative Review

Enforcement. Consequences.

Risk scoring. 3-year review cycle. Fiscal workbook FA1–FA4 (moved here from Accounting — these are enforcement consequences, not financial instruments). Serious Deficiency. School Reviews. Technical Assistance.

Eligibility & Direct Certification

Who is eligible. Legal determination.

DC matching (SNAP/Medicaid/TANF/FDPIR). DirectVer (Indiana) — sponsor-facing student search and application list upload; Indiana's 5-class model is the deepest DC implementation in the LINQ portfolio and is the reference design for this domain. FRPL. Verification. ISP pipeline → CEP. Core2_DirectCertification* (98) + legacy dbo (49).

Tier 2 · Foundational: Constrain inputs to authoritative domains; never decide outcomes

who, what, where before any program work happens

Identity & Access Management

Auth, roles, SSO, OIDC

CnpObjects rights model. Multi-tenant ICnpContext (resolves GlobalDb + StateDb connections). Harmony auth OIDC. ApiAuthenticationTokens. Password policy. JumpSessions (impersonation audit).

Configuration & Tenant Management

State variation lives here — never in code branches

Feature flags (LaunchDarkly). CnpDefaults. Program year rules. ClaimRateAmounts (effective-dated — annual rate updates are INSERTs here, not deployments). Tenant provisioning.

Site & Sponsor Management

Authoritative record of who participates

Sponsors, SFAs, sites, providers, addresses. NDL verification. Agreement number generation. Associates (FSMC-linked). Program enrollment linkage to Program Registry.

Applications & Agreements

Enrollment & eligibility to participate.

Application packets. Sponsor/site/provider applications. Annual Forms framework (AnnualFormTypes, AnnualFormSessions). Annual rollover engine. Board members. Pre-screen. Potential sponsor pipeline.

Tier 3 · Operational: Own program rules & integration surfaces

PROGRAM DOMAINS: OWN PROGRAM RULES & CLAIM LAWS

SNP School Nutrition

NSLP / SBP rules. SSO / SSFO (non-congregate PY2024+). SMP. PLE. Meal pattern compliance. Food safety. Professional standards. October survey. Provision 2/3 ratios.

CACFP Child & Adult Care

Center program rules. VCA Checklist (16-table scoring module). Institution management plans. ACQR reporting. Center budgets. DCH review track.

DCH Day Care Home

First-class program domain — not embedded in CACFP. Provider applications. Provider claims. DCH sponsoring org. Tier I/II rates. 23-day max operating days. License expiry gate.

SFSP Summer Food

Summer site rules. Non-congregate service — 3 modes: congregate / non-cong / mixed. Rural & heat waiver (Healthy Meals Healthy Kids Act 2022). 2% second-meal rule (sponsor-level calculation).

FFVP Fresh Fruit & Veg

Grant-based (not reimbursement-based). Site invitations. Grant allocations. Monthly purchase claims. FFVP Application workflow (Maine reference impl) — dedicated application packet separate from eligibility and entry. First-class domain — not embedded in SNP.

CEP Community Eligibility

ISP × 1.6 = claiming % (capped 100%). CEP groups + sites. 4-year cycle. Dedicated REST API bounded context — not embedded in SNP. DIS CnpCepApi already proved this boundary.

SSFO Seamless Summer

Distinct ClaimLaws and site application rules from NSLP. Non-congregate applies here (PY2024+). ~16 dedicated tables. Architect's call — correct separation.Distinct ClaimLaws and site application rules from NSLP. Non-congregate applies here (PY2024+). ~16 dedicated tables. Architect's call — correct separation.

MODULES: BUSINESS-CRITICAL OPERATIONAL MODULES

VCA Checklist

16 tables. Viability/Capability/Accountability self-assessment. VCAScoringTools definitions in cnp_global (shared all states). Sponsor instances in cnp_ops.

Budget Module

Budgets, Budget Details, Budget Items, Budget Labor Details, Budget Meal Worksheets. Ohio extends to 42 tables in cnp_ext.

Contracts & Procurement

FSMC contracts. Written Approval Requests. Capital Expenditure Requests. Federal Award Letters. Procurement review track. Architect's call — correct separation from Applications.

Risk Management

Risk scoring. Review prioritization dashboard. Reports. Feeds Compliance review scheduling — does not decide enforcement outcomes. Stays specialist.

Training & Tech Assist

Professional Standards (USDA-mandated). Course catalog in cnp_global. TA codes, areas, items, focuses. HI is only DIS state with this built.

⚠ Food Distribution

128–304 tables embedded in shared dbo — cannot be cleanly separated without schema surgery. Ohio/WY have fdp/gratefdp schemas. Decision required before Phase 3.

Waivers & Approvals

Waiver Items, Waiver Types. Written Approval Requests (USDA exceptions). Capital Expenditure Requests. Linked to applications and claims.

PLATFORM SERVICES: SHARED INTEGRATION SURFACES

Forms Engine

Year-versioned definitions in cnp_global. Metadata-driven rendering. AnnualFormTypes, ReviewVersions, ReviewQuestions. Replaces forked versioned ASPX pages.

Integration & Data Exchange

IDcFeedAdapter · IAccountingExporter · ILicenseAdapter · IEdFiAdapter. USDA ArcGIS. NDL. LaunchDarkly Session Replay. Format translation only.

Workflow & Task Management

Hangfire background jobs. FFY rollover automation. CnpTasks. Review scheduling calendar. CTM replacement

Communications & Notifications

Email templates (cnp_global). MessageBoards. BroadcastEmail. Intercom in-app messaging. System alerts. Outbound only — no business rules.

Document Library

File storage. Form PDF snapshots → S3. Document versioning. DownloadFormBlobs. Pre-signed URL generation for historical form access.

PER=STATE ADAPTER PATTERN

Addresses failure mode #7 — "state-specific behavior had no clean home." Every state-specific behavior lives in one of these four adapter slots. No state-keyed if/else branches anywhere in shared platform code.

IAccountingExporter

UT→USAS · CO→COFRS · SC→CALSTARS · MT→SABHRS · TN→ISAS · OH→SCEIS

IDcFeedAdapter

Per-tenant DC source config: SNAP, Medicaid, TANF, FDPIR. PGP/SFTP or REST per state.

IEdFiAdapter

States with Ed-Fi ODS: RestEdFiAdapter. States without: FlatFileEdFiAdapter fallback.

ILicenseAdapter

UT→DOPL license sync. Per-state childcare provider license verification.

Tier 4 · Observational: Read models & outbound services; never influence decisions above

Reporting & Analytics (snowflake)

Read models only

ALL reporting queries run against Snowflake REPORTING layer — never against Aurora cnp_ops. FNS-10/44/418/640/905 pre-materialized nightly. Zero operational DB contention. Tenant isolation via Snowflake row access policies. Replaces Architect's Reporting & Data Export domain as an application concern.

Audit & Observability

Event consumption only

Never influences behavior. Audit trail for eligibility, claims, payments, config. Activity logs. Datadog APM. User access reports.

Notifications & Communications

Outbound only

No business rules. Email templates. Broadcast messages. Notices. In-app messaging (Intercom). Discussion board. System alerts.

Reference Data & Transparency

Lookups — no decisions

Program codes, county types, poverty data, USDA calendar, FY tables. Public AR summary portal (feature-flagged per tenant).

Database Architecture

DATA SPINE

ProgramGroups → ProgramGroupYears → ProgramGroupEnrollments

Every Application and Claim FK-chains through **ProgramGroupYears**. It's the structural spine of the data model. · Definitions in cnp_global · Enrollment records in cnp_ops.

CNP_GLOBAL 1 shared instance

Contents: ~387 tables: ProgramMealRates, ClaimRateAmounts (effective-dated), all *Types/*Codes/*Statuses, ReviewVersions,

CNP_OPS 1 instance per state

Contents: ~1,015 tables: all Sponsors, Sites, Applications, Claims, Reviews,

ReviewQuestions, VCAScoringTools, AnnualFormTypes, BudgetItems, TrainingCourses

Update model: Annual rates = INSERT with EffectiveDate. Zero deployment. Zero SP change. Zero per-state work.

Access: Read-only for all state instances. ICnpReferenceCache (24hr TTL for types, 1hr for rates).

Disbursements, Users, DC records, VCA instances, Budget instances

Isolation: Database IS the tenant boundary. No TenantId column needed in operational tables. State data sovereignty guaranteed at infrastructure level.

Extensions: cnp_ext schema for state-specific tables (Ohio Budget+42, GA legacy tables). Never in shared platform code.

State Variation Handling

No domain anywhere contains a branch on which state it's running. Every state difference is a flag, an adapter, a config value, or a separate extension schema.

01 Feature Flags

Launch Darkly

Whole capabilities toggle per tenant — AR transparency portal, non-congregate SFSP, state-specific report packs. One codebase, many footprints.

02 Plug-in Adapters

4 interfaces only

IAccountingExporter (UT→USAS · CO→COFRS · SC→CALSTARS · MT→SABHRS · TN→ISAS · OH→SCEIS). IDcFeedAdapter (SNAP/Medicaid/TANF/FDPIR per tenant). IEdFiAdapter. ILicenseAdapter. Cnp.Extensions.<State>.dll resolved at startup.

03 Configuration

JSON B | cnp_global

State-specific values — claim rate overrides, deadlines, program-year rules, CnpDefaults — stored as effective-dated rows in cnp_global config schema. No schema

04 Extension Schema

cnp_ext per state

Genuinely unique state needs live in cnp_ext schema within the same Aurora cluster. Ohio's 42-table budget extension, Georgia's

forking. Readable at runtime, no deployment.

legacy tables, Iowa's fnm schema. Never bleeds into shared platform code.

Appendix

DOMAIN ORIGINS

Added references to DIS items we should include in future State CNP Platform

Domain / Decision	Source	Rationale
Program Registry (~7 tables)	Combined_v3	"Program Registry" is sharper than "Program Enrollment Model"
SSFO as own program domain (~16 tables)	Combined_v3	Distinct ClaimLaws and site application rules from NSLP
CEP as dedicated bounded context	New	DIS CnpCepApi proved the boundary works; ISP math needs isolation
Contracts & Procurement (~57 tables)	Combined_v3	Distinct enough from Applications; FSMC procurement is a real workstream
Training & **Technical Assistance** combined (~57 tables)	Combined_v3	Same stakeholders, same lifecycle — grouping is correct
Fiscal Actions → Compliance, not Accounting	New	FA1–FA4 are enforcement consequences, not financial instruments
Reporting → Snowflake (observational, not operational)	New	Reflects database-per-tenant and Snowflake decisions
Authority tiers	New	Determines service dependency rules

ClaimLaws mechanism callout	New	Directly addresses failure mode #5 (no SP characterization)
cnp_global / cnp_ops database split	New	Reflects database-per-tenant decision
VCA Checklist, Budget, Waivers	Schema analysis	Confirmed in all 5 CTS state schemas
Commodities scope gate	Both	128–304 tables embedded in shared dbo — cannot be cleanly separated